

02_HARDWARE_INTEGRATION_RESULTS

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EVIDENCE PROVENANCE: [PHYSICAL (BENCH)]
Measured optical tachometer frequency and IMU tilt angle responses on dual ESP32 physical test benches.

1. Optical Tachometer Staircase Response

PWM Duty	Vehicle A Speed	Vehicle B Speed	Encoder Pulse Rate	Verification
0	0.00 m/s	0.00 m/s	0 pps	Zero boot movement verified
60	0.28 m/s	0.24 m/s	51.4 pps	Linear engagement
120	0.68 m/s	0.62 m/s	124.7 pps	Mid-range cruising
180	1.12 m/s	1.02 m/s	205.4 pps	Fast haul speed
220	1.40 m/s (MAX)	1.21 m/s	256.8 pps	Vehicle A maximum ceiling
240	--	1.30 m/s (MAX)	238.5 pps	Vehicle B maximum ceiling

2. IMU Inclination Measurement on Ramps

Calibrated tilt test blocks verified MPU6050 pitch tracking: 8.0% grade (4.57 deg) yielded 4.52 deg (+/- 0.11 deg); 12.0% grade (6.84 deg) yielded 6.79 deg (+/- 0.14 deg).